

Trier University
Faculty VI - Regional and Environmental Sciences
Department of Environmental Remote Sensing and Geoinformatics

Forest disturbance monitoring and anomaly detection within the Hunsrück-Hochwald National Park

Portfolio for
ECOSYSTEM REMOTE SENSING
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Lecturer: *Dr. rer. nat. Johannes STOFFELS*

Patrick CHRISTEN 1506810 s6pachri@uni-trier.de	Karel Loic NIMPA FOKOUA 1734880 s6kanimp@uni-trier.de
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1. Introduction

Forests play a vital role in the environment, regulating climate systems and combating erosion (Grossiord et al., 2015). They also provide several ecosystem services (wood production, medicinal plants, etc.) and fulfil several social and cultural functions (Bahuchet, 1992). However, forest ecosystems are under increasing threat from various factors, including forest degradation, human activities and climate change, which accentuates attacks from insects such as spruce bark beetles which tends to spread rapidly under extreme meteorological effects. This leads to the ecosystem disruption and significant biodiversity loss (Seidl et al., 2017).

In Europe, forest cover is around 33% of the land area and their management is essential for conserving biodiversity and mitigating climate change (European Commission, 2023). Unfortunately, European Spruce bark beetle (SBB) is one of the biggest threats of central European forests and can therefore cause forest dieback (Hallas et al., 2020). Because of climate change and the increased vulnerability of trees, Hunsrück-Hochwald National Park (HHNP) is not an exception to this phenomenon. This forest ecosystem is heavily dominated by coniferous trees, particularly spruce (*Picea abies*), which play an important role in habitat structure and ecosystem dynamics (Heinrichs et al., 2019). However, in recent decades, the park's spruce forests have come under increasing pressure from bark beetle infestations. Once they have infested a tree, they destroy the xylem and phloem of the tree, rapidly leading to red needles followed by the death (Biedermann et al., 2019).

Previous work has already demonstrated the effectiveness of remote sensing in monitoring forest health through indices (Rodríguez Paulino et al., 2024). In this context, remote sensing offers an effective way to assess the impact of SBB on a large scale. It enables affected areas to be mapped and early signs of forest stress to be analysed. However, detection of green attacks remains difficult (Huo et al., 2023).

2. Research question and objectives

This portfolio therefore, will try to answer the main question: how beetles affect the forest ecosystem of HHNP? This question leads to two main objectives: map forest loss caused by

bark beetle attacks in the Hunsrück National, then assess the state of health of coniferous trees by analysing signs of decline gathered from a field survey and then compare it to satellite imagery-based maps. To achieve the objectives, the 2024 forest loss maps of HHNP will be produced, then the health status of coniferous trees will be mapped and analysed based on survey data.

3. Study Area and Datasets

3.1. Location of study area

Since Germany has few national parks, the states of Rhineland-Pfalz and Saarland established the HHNP in 2015 with the aim of increasing and preserving biodiversity, thereby restricting the forest diversity it contains to a single ecological and conservation use (König et al., 2016). According to König et al. (2016), the HHNP is located in the south-west of Germany and is shared between the states of Rhineland-Pfalz and Saarland. It covers an area of around 102 km² and consists mainly of forest.

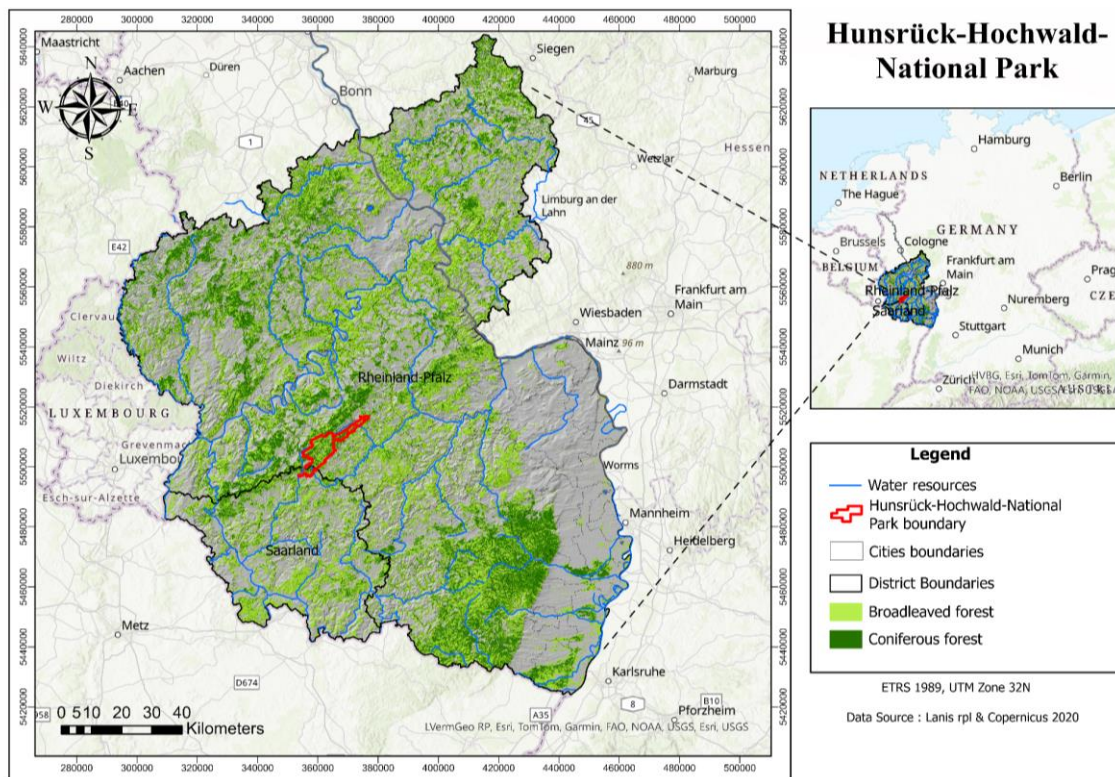


Fig. 1: Overview of the study area

The HHNP is divided into 3 specific zones, wilderness area (zone 1a), development area (zone 1b) and maintenance area (zone 2). In addition, it also contains special conservation areas (ZSC) as defined by the Natura 2000 programme (Naturschutzverwaltung Rheinland-Pfalz, 2020).

3.2. The Physical environment of HHNP

- **Relief and geology**

The relief in HHNP consists of several mountain ranges that extend in a southwest-northeast direction. The park's highest elevation is the Erbeskopf with 816.3 metres of altitude; the park is around 32 km long and between 2 and 8 km wide (König et al., 2016). Regarding the geology of the HHNP, the bedrock includes shales, quartzite, and sandstone. These rock layers are strongly influenced by folding, which gives the region its characteristic appearance (Knautz, 1992). The soil structure is shaped by a long history of weathering, as a result the highest slopes are covered with quartzite debris. Soils in these areas are generally coarse-grained, shallow and sandy, with a mixture of raw soils, rankers, podzols and brown earths (Knautz, 1992).

- **Climate**

The national park's climate is predominantly oceanic, with heavy rainfall in winter. However, the valleys within the park can experience more continental conditions. Klauck (1985) highlights that rainfall levels vary from around 800 mm in the lower valleys to over 1,300 mm per year in the high mountain areas. According to him, the main winds blow from south to west, contributing to the higher levels of precipitation on the south-western slopes of the park. The number of days of snow cover is also highly variable, ranging from 80 days in the highest areas to 30 days in the lowest parts.

- **Fauna and Flora resources**

The national park is heavily wooded, with over 98% of its surface area covered by woodland, 24% of which consists of very old trees. The park enjoys a privileged status due to the presence of very old forests. Despite this, the natural structure of the forest has changed considerably, with conifers now accounting for a large proportion of the woodland. (König

et al., 2016). Heinrichs et al. (2018) indicate that beech forest is the dominant potential natural vegetation (PNV); below the altitude of 500-520 m, the PVN is a mixed Luzulo-Fagetum forest. However, as a result of human activity, the composition of the forest has changed significantly, replacing much of the natural deciduous forest with plantations of conifers, mainly Norway spruce (König et al., 2016). Aside from forests, the park also contains around 1,300 hectares of peat bogs and wetlands. König et al. (2016) suggested that these peat bogs are home to species such as marsh birch and various sphagnum mosses.

3.3. Data acquisition

To achieve the objectives of this portfolio, data has been collected in various databases. These are mainly the GIS databases for the Rhineland-Palatinate and the Saarland. Data was also collected from various Sentinel-2 satellite images, OpenStreetMap (OSM) and from data collected in the HHNP.

4. Methodology

The methodology used in this analysis varied according to the research objectives.

4.1. Forest loss mapping

In order to map forest loss, a number of steps were taken, from identifying the forest areas of the park to mapping the types of forest and forest disturbance.

4.1.1 Forest / Non-Forest Map

To map forest and non-forest area, the forest layer from OSM was used as a basic mask to specify the region of interest. Then, 14 Sentinel-2 scenes were combined into a layer stack of 140 bands. Following that, a principal component analysis (PCA) was run using only the Green, NIR and SWIR bands from the stack (These bands were selected to overcome memory limitations when trying with all 10 bands). As a result, of the 42 principal components (PC), 10 were retained, explaining 90 % of the variance. The first PC (PC1) shows cloudy areas and is thus removed from the unsupervised K-means classification, which is performed on PC2 - PC10. The suggested clusters are then assigned to the 'Forest' and 'Non-Forest' thematic classes.

4.1.2. Forest Type Mapping

The forest typology has been carried out using the forest layer from OSM as a ROI mask. This mask was combined with the non-forest areas from the K-means classification. About 2000 random selected points were created to visually check the forest type (deciduous or coniferous). As a result, 1195 points were retained. The points were split into training (75%, 896 points) and validation points (25%, 299 points). Then, using training points, a Random Forest Classifier (RFC) with 50 trees, was applied to the April scene of Sentinel-2 in two different cases:

case 1: use all 10 Sentinel-2 bands

case 2: use all 10 Sentinel-2 bands + Copernicus 30m global DEM

The goal of case 2 is to compare the utility of topographic information in the forest type and non-forest classification. The April scene has been used with the goal of improving the differentiation between deciduous trees and conifers; the conifers should generally be evergreen while the deciduous trees should yet be leafless.

Also, the training points were used as independent validation data for the K-means classification. An accuracy assessment has been made using an area adjusted confusion matrix according to Olofsson et al (2013). Using the validation points, values from the classification for case 1 and case 2 were extracted, followed by a removal of NULL-values. This resulted in 203 remaining points for the validation.

4.1.3. Mapping of forest disturbance

As the study was focused only on coniferous forest, we reclassify the forest type map to create a coniferous mask. Then, NDVI and PVI were calculated from the sentinel 2 scene with the same acquisition period (13/06/2023 and 25/06/2024) to compare changes using the index. June scenes have been selected to specify the peak reflectance for conifers. Following that, the index of both acquisition periods has been calculated by extracting index bands and the proportional difference of the index value has been calculated using formula 1.

$$\text{Proportional difference} = (\text{Index}_{2023} - \text{index}_{2024}) / \text{Index}_{2023} \quad (1)$$

The mask of coniferous trees for 2023 was applied to only keep the proportional difference of coniferous areas. Next a z-transformation was applied to the proportional difference to set the mean at 0 and changing the measurement to multiples of the standard deviation of the samples using formula 2.

$$z - transformation = (Proportional\ difference - mean) / sigma \quad (2)$$

Afterwards, the proportional differences were reclassified into ranges specified in Tab. 1.

Tab. 1: Classes assigned for the reclassification of the z-transformed proportional difference including the assigned ranges.

Class name	Value range
regrowth	] -inf; -2]
slight regrowth	] -2; -1]
no change	] -1; 0]
slight loss	] 0; 1]
loss	] 1; 2]

In a post-processing step, OSM roads were removed with a 10 m buffer and finally the boundary clean tool was used to smoothen class boundaries and to clump the classified output. Finally, the descending sorting was used to sort the zones by size in descending order, where the larger zones are then prioritised to expand into smaller zones (ESRI, n.d.).

4.2. Health state of coniferous trees

To examine the health of the coniferous trees, samples were collected in the study area to gather information from the three management zones of the park, then this is compared with various maps produced from satellite data.

4.2.1. Sampling design

To achieve this, a grid of cells based on the Landsat resolution of 30 m was constructed inside the park. Then, on the basis of a forest typology map, areas rich in non-coniferous trees were excluded. In order to analyse the impact of the different national park zones, an area of 1x1 km² was preselected, where all 3 zones are similarly included. At the same time, the different areas were meant to be distributed to cover the majority of the HHNP. They should also be in close proximity to free parking-spaces, to guarantee quick accessibility of the areas. Within each area, we have applied a random selection of about 30 points per zone with a minimum distance of 90 m between points of the same zone. The intention of this is to reduce spatial autocorrelation but still account for a potential heterogeneity in the field, associated with the different managements in the zones.

Finally, a point has been generated in the centre of each cell. The following unique cell feature identifiers (CFID) have been assigned to each point. This CFID consists of a continuous number followed by the designated area (0001a1 = the first survey point of area 1). The numbering order has been fixed to have following numbers in close proximity, in order to separate each point easily from other groups: To end this process, a map has been produced including the survey points with high resolution satellite imagery as a base-map, including a basic road network and location labels.

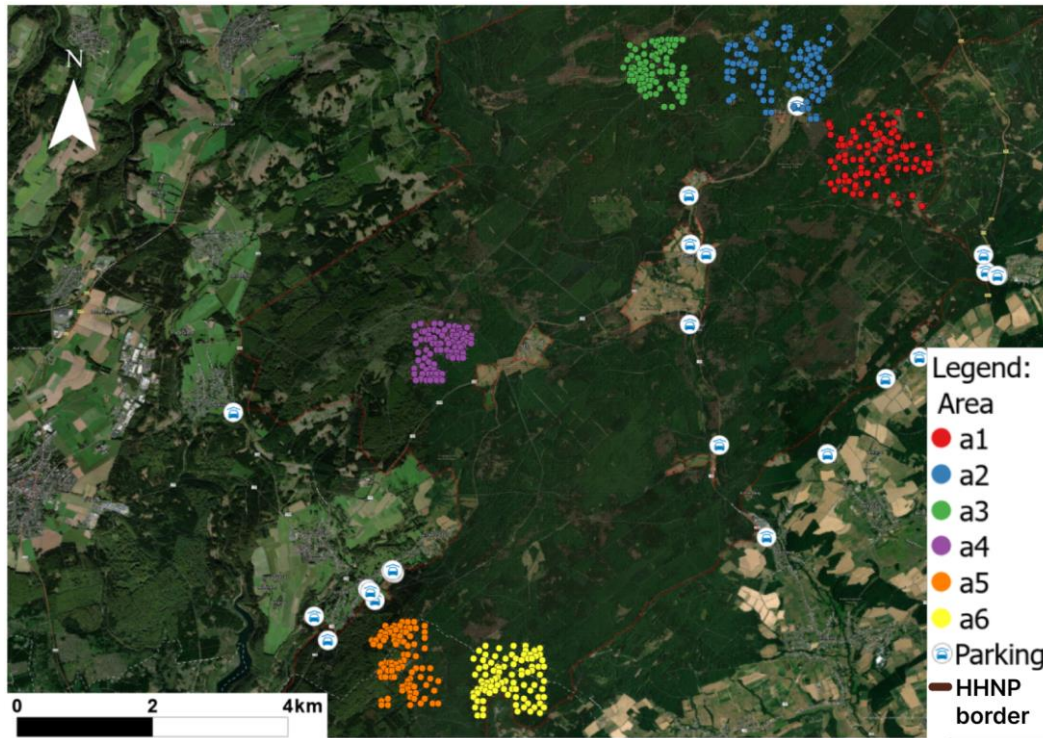


Fig. 2: Survey points according to their area in the HHNP.

4.2.2. Data collection

Data collection in the field was done based on sample plots. In each survey area, circular plots with a diameter of 20 metres were set up. The centre of the plot corresponds to the centre of the survey cell, and from the centre of the plot, we have measured 10 m in the directions of North, South, East, and West while setting reference markers. Inside the plot, data collection was focused on the dominant species, forest type (deciduous, coniferous, dead forest) and the signs of sickness (red needles, loss of bark, green needles on the ground). Canopy cover was also taken into consideration, but it was evaluated visually and a photo of the canopy from the ground was saved to recheck the canopy cover.

5. Results

This part of the study presents all the results obtained with a view to facilitating their discussion

5.1. Forest loss mapping

5.1.1. Sensitivity NDVI vs. PVI

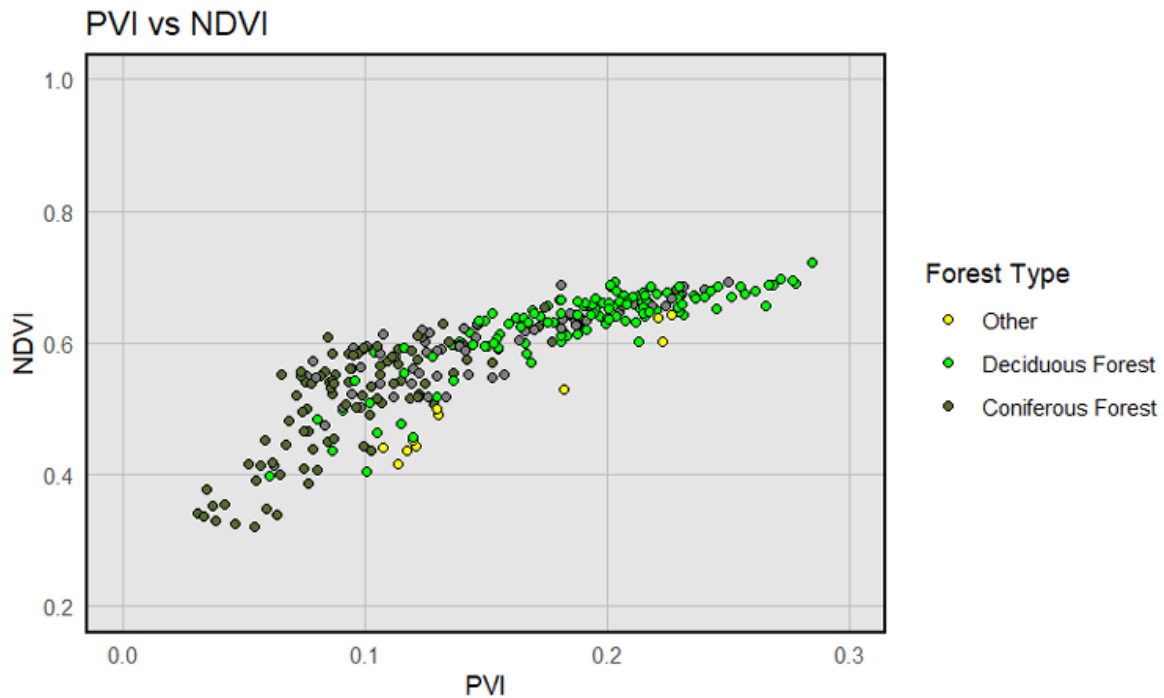


Fig. 3: Scatterplot of NDVI vs. PVI for various forest types from the Copernicus Forest Type Layer, from Sentinel-2 scene dated 27/05/2023

The scatter plot extracted from Sentinel-2 imagery, making use of the Copernicus Forest Type Layer, shows that deciduous forest values are generally between $\sim 0.6 - 0.7$. A saturation effect in NDVI values at ~ 0.7 , marking a pseudo-plateau, which is particularly evident in values extracted from deciduous forest, is observed. Most of the PVI values for deciduous forest are in the range of $\sim 0.15 - \sim 0.3$. Here, no pseudo-plateau can be observed.

5.1.2. Forest / Non-Forest Map

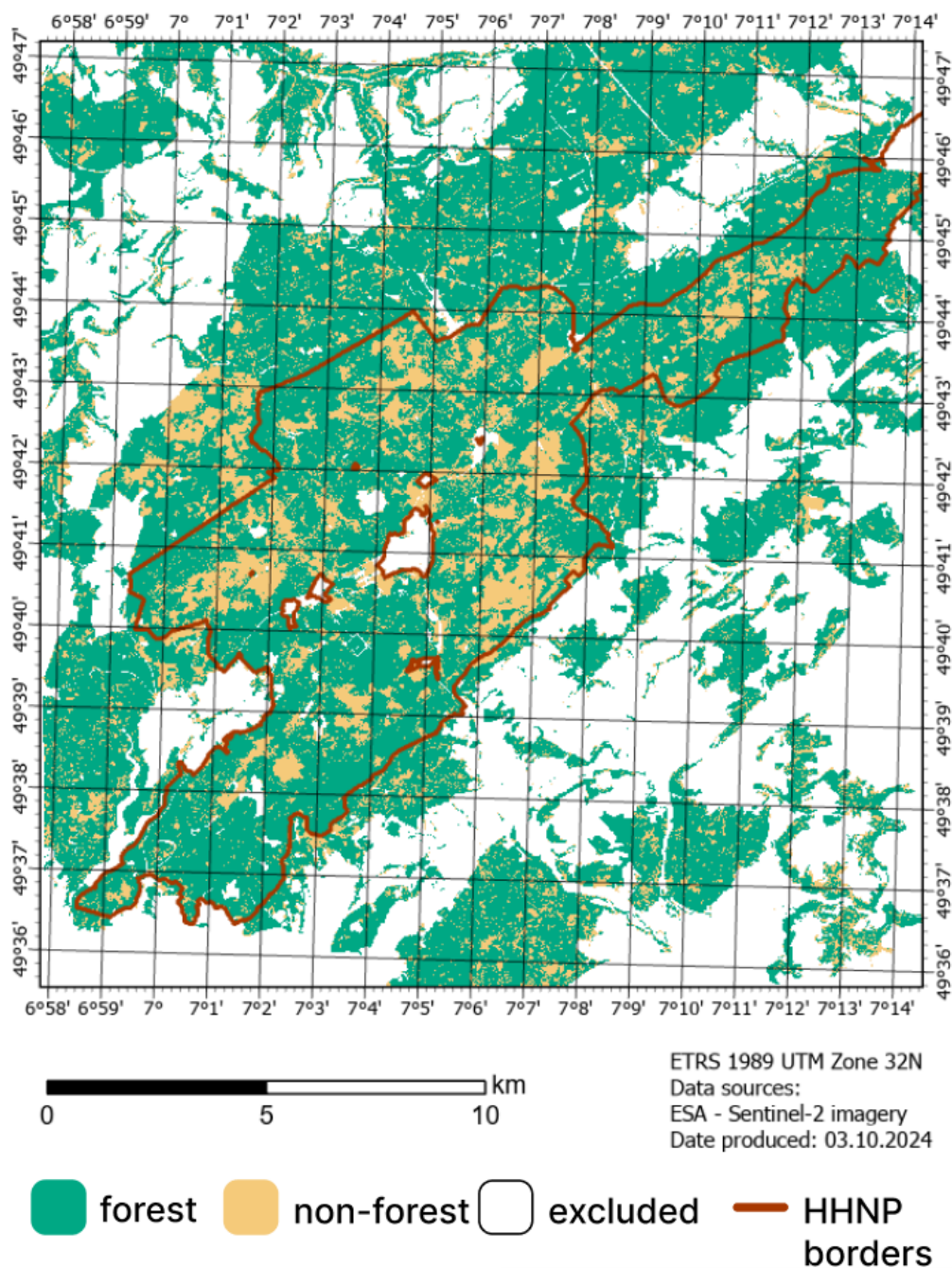


Fig. 4: Forest / Non-Forest map for HHNP 2023.

The result of the forest and non-forest map shows that the park's surface area is dominated by forest. The distribution of these two types of land cover shows a large distribution of non-forest species in the central north-east of the park.

Tab. 2: Confusion matrix (area adjusted) for the Forest - Non-Forest map, including producer's, user's and overall accuracy (in percent) with margin of error (95 % confidence interval).

Confusion matrix (proportional): Forest - Non-Forest							
	ref (area %)		Sum		Accuracy (%)		
class	1	2	Map	Ref.	PA	UA	OA
1	76.44	3.34	79.79	87.49	87.37 +- 1.38	95.81 +- 1.40	85.60 +- 1.78
2	11.05	9.16	20.21	12.51	73.26 +- 7.20	45.33 +- 6.80	

Class 1 = forest, class 2 = non-forest.

Tab. 2 shows the area adjusted confusion matrix including the producer's accuracy (PA), the user's accuracy (UA) and the overall accuracy of the Forest / Non-Forest map (OA). The OA is at 85.60 ± 1.78 %. The PA and UA of the forest class are relatively high with 87.37 ± 1.38 % and 95.81 ± 1.40 % respectively. On the other hand, the non-forest class has a relatively low PA and UA with respectively 73.26 ± 7.20 % and 45.33 ± 6.80 %. Notably, the margins of error for the non-forest class are much larger (f.e. ~ 7 % compared to 1.4 %). As a result, 76 % of the validation points were classified and referenced as forest while the non-forest class only makes out 12.5 % of the validation points.

Tab. 3: Adjusted area estimation for the classes (km²), based on their estimated proportions, including margin of error (95 % confidence interval).

class	proportion	proportion SE	count	area adj.
1	0.8749	0.0174	2237630	245.37 +- 9.76
2	0.1251	0.0174	566886	35.08 +- 9.76

Class 1 = forest, class 2 = non-forest.

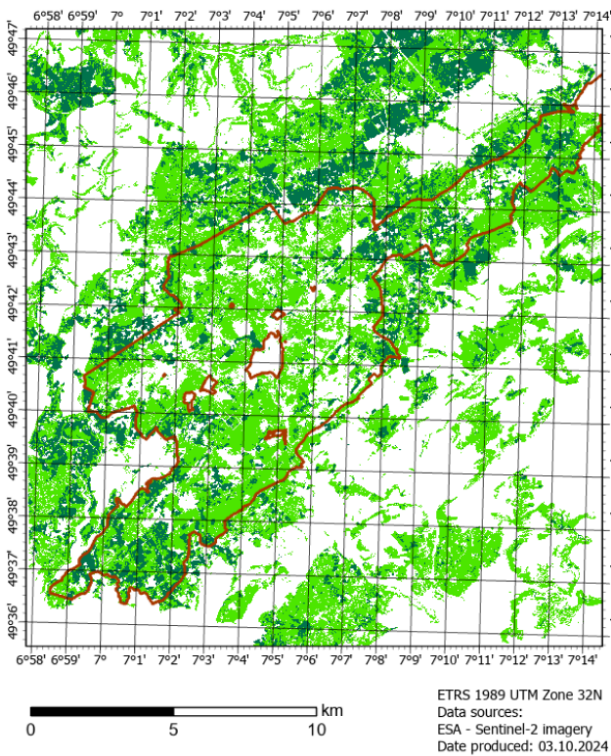
Tab. 3 indicates the proportions of the classes contributing to the entire map. These proportions are then used to calculate the adjusted area estimation, including margins of errors, based on the standard error of the proportions (Olofsson et al., 2014):

*Adjusted area = proportion of class * total error $\pm$ 1.96 * proportion standard error (SE) of class * total area.* (3)

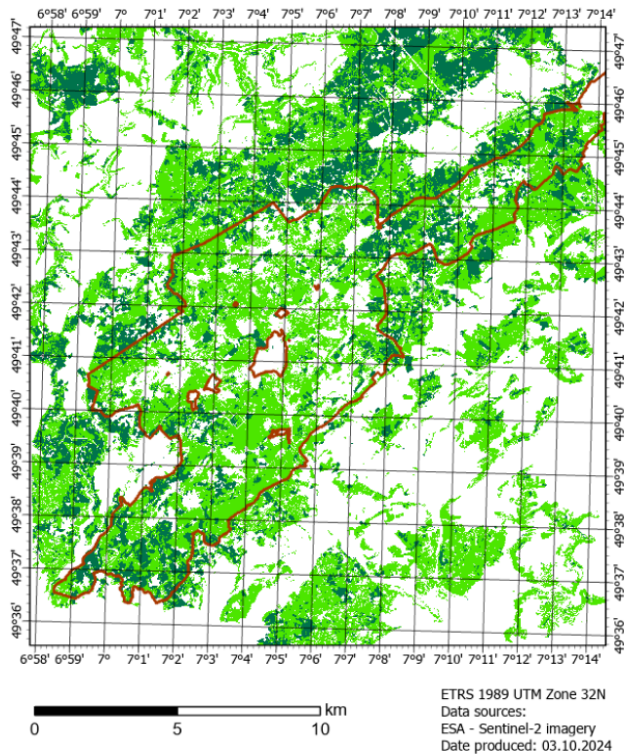
The forest class in total makes up 87.5 %. The adjusted area estimations for the forest class are 245.37 ± 9.76 km² (non-adjusted area = 223.76 km²) and for the non-forest class 35.08 ± 9.76 km² (non-adjusted area = 56.69 km²) for a confidence interval of 95 %.

5.1.3. Forest Type Map

a) case 1:



b) case 2:



forest non-forest excluded HHNP borders

Fig. 5: Forest Type map for HHNP 2023 for cases 1 and 2.

Above are shown two maps displaying the result from the RFC for the cases 1 and 2. For case 1, the highest contributors to the RFC were band 10 followed by bands 3 and 8 while for case 2, such was band 10 followed by bands 6 and 4.

Tab. 4: Confusion matrix (area adjusted) for the Forest type maps of cases 1 and 2, including producer's, user's and overall accuracy (in percent) with margin of error (95 % confidence interval).

Confusion matrix (proportional): Forest Type (case 1)							
	ref (area %)		Sum		Accuracy (%)		
class	1	2	Map	Ref.	PA	UA	OA
1	61.18	11.70	72.88	68.16	89.75 +- 1.95	83.94 +- 3.14	81.31 +- 2.73
2	6.98	20.13	27.11	31.83	63.23 +- 4.84	74.24 +- 5.38	

Confusion matrix (proportional): Forest Type (case 2)							
	ref (area %)		Sum		Accuracy (%)		
class	1	2	Map	Ref.	PA	UA	OA
1	63.92	8.97	72.89	70.61	90.53 +- 1.78	87.69 +- 2.88	84.34 +- 2.52
2	6.69	20.42	27.11	29.39	69.49 +- 5.16	75.34 +- 5.04	

Class 1 = Deciduous, class 2 = Coniferous.

Tab. 4 shows the results from the validation, including the confusion matrices for each case. Case 2 has generally higher accuracies than case 1, with lower margins of error. In both cases, the deciduous forest class shows higher accuracies than the coniferous forest class. The margins of error for deciduous forest are also generally lower than for coniferous forest.

Tab. 5: Adjusted area estimation for the classes (km²) of the Forest Type maps (for cases 1 and 2), based on their estimated proportions, including margin of error (95 % confidence interval).

case 1:				
class	proportion	proportion SE	count	area adj.
1	0.6817	0.0534	1637537	153.16 +- 23.99
2	0.3183	0.0534	609171	71.51 +- 23.99

case 2:				
class	proportion	proportion SE	count	area adj.
1	0.7060	0.0494	1637537	158.62 +- 22.20
2	0.2940	0.0494	609171	66.05 +- 22.20

Class 1 = Deciduous, class 2 = Coniferous.

Tab. 5 shows the adjusted area estimations resulting from the confusion matrices; case 1 results in higher conifer proportions than case 2. Case 1 leads to adjusted area between 129.17 - 1777.15 km² for deciduous forests and 47.52 - 95.5 km² for conifers, while for case 2 deciduous forests are between 136.42 - 180.82 km² and conifers between 43.85 - 88.25 km².

5.1.4. Mapping of forest disturbance

Forest disturbance mapping shows areas with canopy loss, mostly located in the southwest and northeast of the park, as can be seen in Fig. 6.

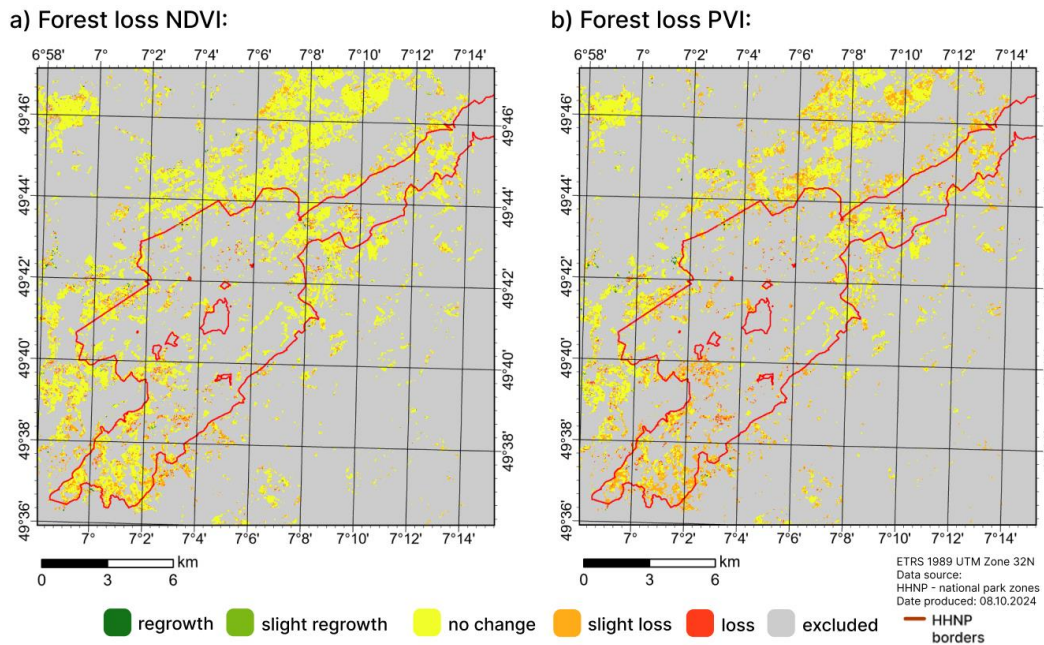


Fig. 6: Overview of the forest loss (FL) maps based on a) the NDVI and b) the PVI, using Sentinel-2 scenes from 2023 and 2024 as input.

For the entirety of the HHNP we can see some variance in the spatial distribution of the forest change. Outside of the national park, most of the conifer areas experience no change (see NW and SW around the HHNP in Fig. 6). This is also the case for SE of the NDVI - product. More areas indicating a slight loss can be observed in the centre and south within the national park, as well as NE above the HHNP for the PVI - product. Generally, the “no change” class is represented the most, while the PVI - product displays more “slight loss”. Regrowth is barely included in the area of interest.

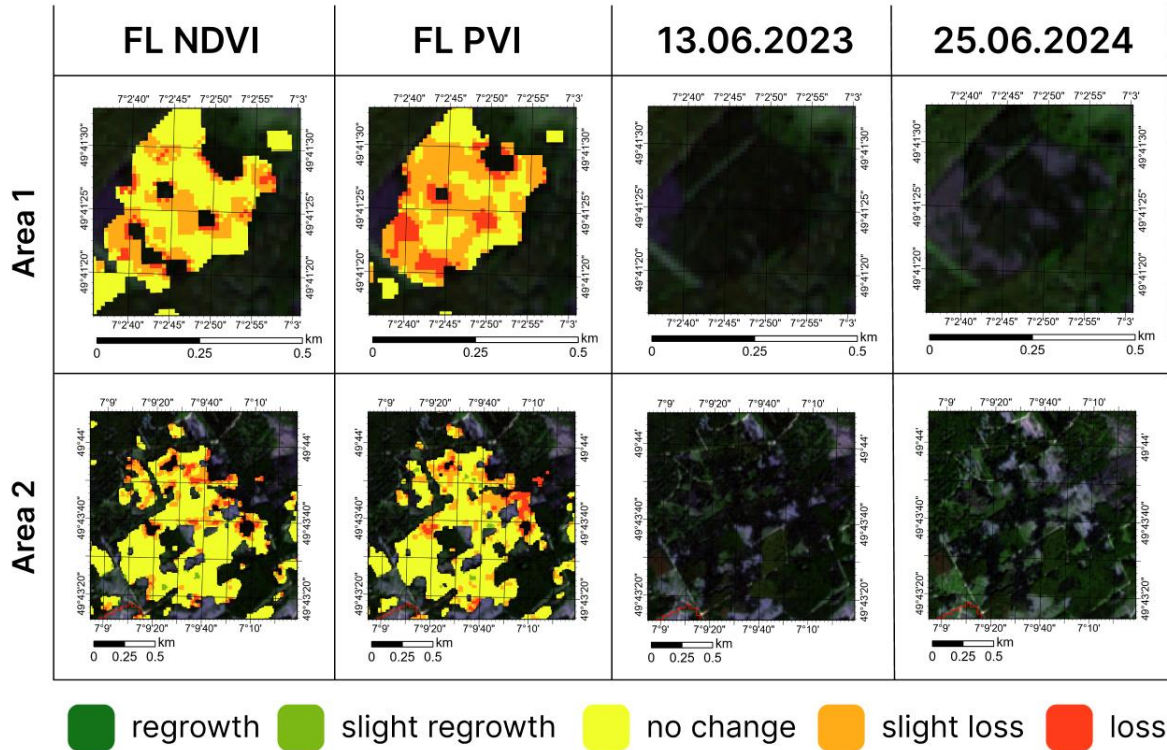


Fig. 7: Two close-up areas from forest loss (FL) maps based on the NDVI and PVI, compared to their input Sentinel-2 scenes for 2023 and 2024 as true-colour-images.

A detailed view of some representative areas for the FL maps shows the output for the NDVI - and for the PVI - product side by side with their input Sentinel-2 scenes for 2023 and 2024 (Fig. 7). From a visual inspection, forest loss is detected well and the conifer - to deciduous forest - borders are traced well. But some visually identifiable conifer areas are not included in the maps. The displayed area for the FL PVI is generally larger than for the FL NDVI and there is generally a larger amount of area representing forest loss in the PVI-based product.

5.2. The health state categories of HHNP

After the survey, in total 196 survey points were acquired over the sampled area. From the collected points, 56 points correspond to dead standing trees, 33 are assessed as healthy conifers, 38 as moderately damaged conifers and 13 as severely damaged, totalling 140 points for conifers as shown in Fig. 8.

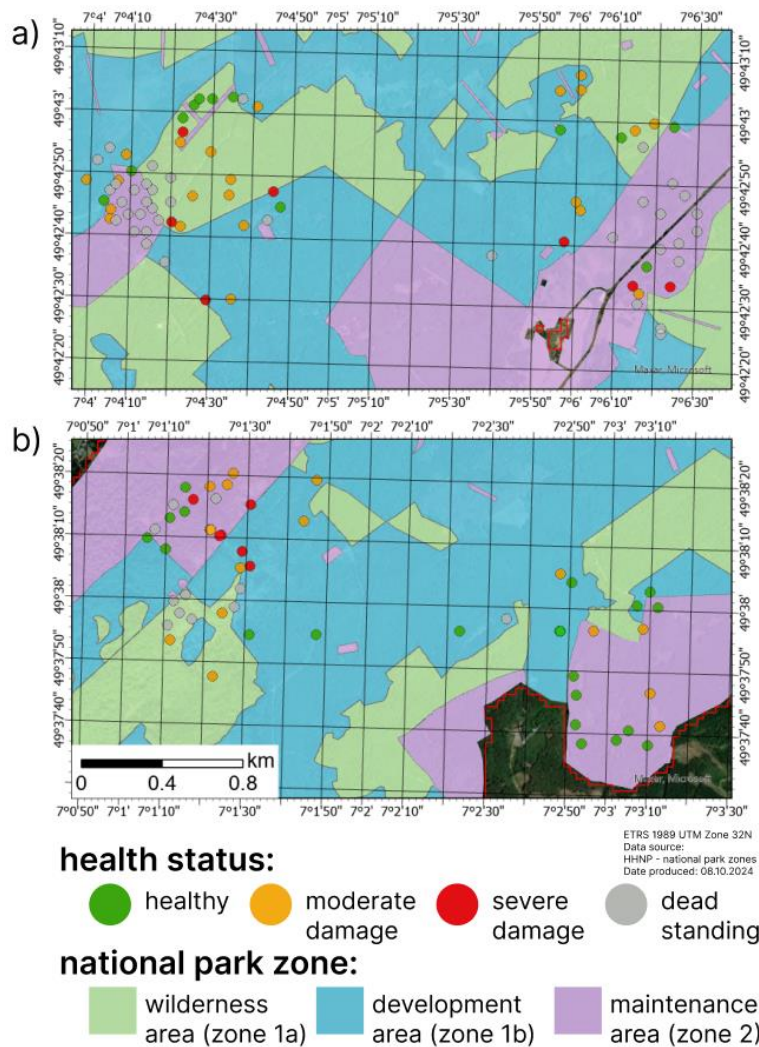


Fig. 8: Overview of survey points for conifers of different health statuses with the respective national park zones they belong to. a) includes area a2 (right) and a3 (left), while b) includes area a5 (left) and a6 (right).

The distribution of the sampled points in the HHNP shows that of the 140 points, 35 points are in zone 1a, 42 in zone 1b and 61 in zone 2. the count and percentage of the different health status categories for each natural park zone (Fig. 9) shows that the severe damage category generally constitutes the least percentage of the observations in each zone. For zones 1a and 2, the dead standing trees make up the majority with ~ 45 %. While for zone 1a, the moderate damage plots are more frequent than healthy plots, both categories are relatively equal in frequency for 2a. Zone 1b shows relatively equal frequencies for all categories except for severe damage.

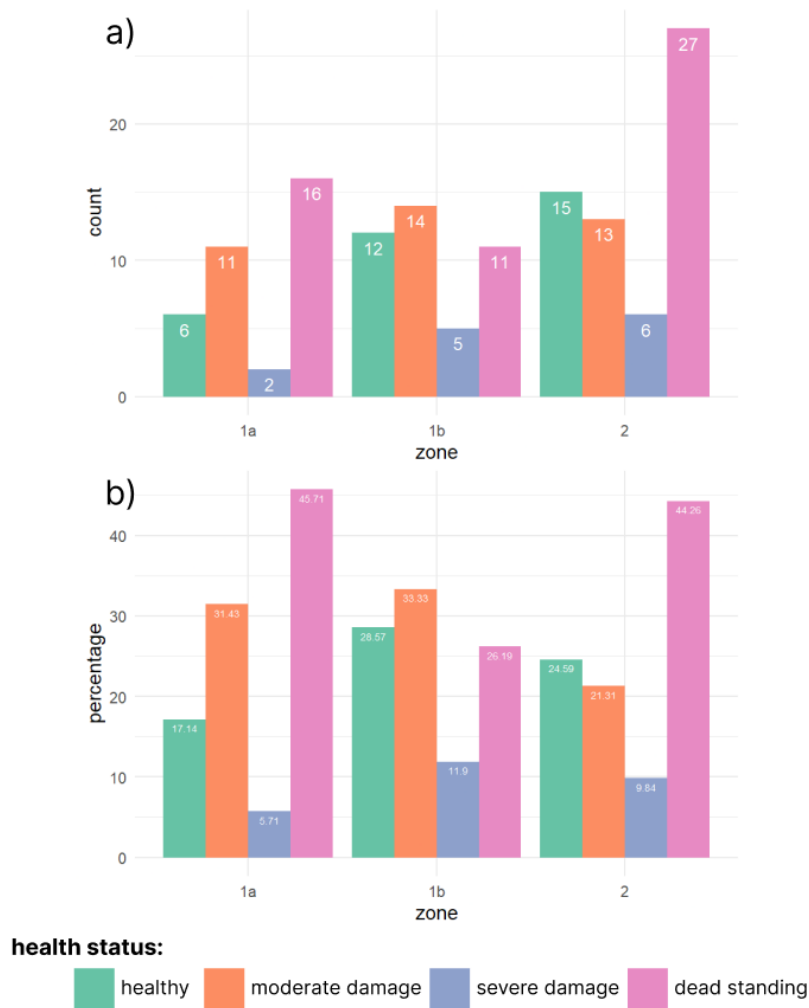


Fig. 9: Barplots indicating a) the count of the different health status categories and b) the percentage of the different health status categories per natural park zone.

With the severe damage category ranging 2 - 6 for the different zones, a pair-wise Fisher's exact test is used instead of the Chi-square test to assess the relationship between the health status and the natural park zone: H_0 states that there is no relationship between the variables. The result from the pairwise two-sided Fisher's test (Tab. 6) with all insignificant p-values indicate that the null hypothesis is not rejected.

Tab. 6: Pairwise two-sided Fisher’s exact test indicating the paired variables (natural park zones) and the associated p-value.

Fisher's exact test	
variables	p-value
1a ~ 1b	0.2704
1a ~ 2	0.6137
1b ~ 2	0.2637

6. Discussion

6.1. Sensitivity NDVI vs. PVI

The saturation in NDVI occurs because the NDVI reaches an upper limit, beyond which it cannot accurately capture additional increases in biomass. Aklilu Tesfaye & Gessesse Awoke (2021) associate this to a maximum of absorption in the red light being reached, once the canopy cover reaches 100%. In contrast, NIR reflectance continuously increases with higher LAI (Tucker, 1977). This results in an imbalance between the slight decrease in the red band with a strong increase in the NIR, leading to little change in the NDVI ratio (Aklilu Tesfaye & Gessesse Awoke, 2021). This effect is especially noticeable in dense forests, such as deciduous forests during their growing season, which is consistent with the data collected on May 27. In contrast, the PVI does not exhibit this saturation effect, allowing for clearer differentiation between forest types and varying biomass levels.

6.2. Forest / Non-Forest Map

The UA indicates a low reliability for the non-forest classification, meaning that about 65 % of the non-forest areas on the map are actually forest. This indicates an over-representation of the non-forest areas on the map. This is underlined by comparing the non-area-adjusted area estimation (56.7 %) to the area-adjusted-estimation (35.1 ± 9.8 %). This would mean that the overestimation should be between 11.9 to 31.4 %. The PA of the forest class further indicates said overestimation. The OA is still relatively high due to the strong influence of the more accurate classification of the forest class.

6.3. Forest Type Map

The general increase in accuracy (Tab. 4) from case 1 to case 2 shows that using topographic information in addition to the Sentinel-2 bands results in a better classification of the forest types. The observed differences are still relatively low, except for an improved PA for coniferous forests and an improved UA for deciduous forests. The improvement due to the topology could be explained by conifers being more prevalent in areas with higher slopes, as they are more adapted to poor conditions and chronic stresses, such as nutrient poverty in pronounced slopes, with deciduous trees dominant under more forgiving conditions (Lusk, 2011). As the accuracy from case 2 is higher, the estimated area intervals from case 2 should be more reliable.

The assumption that conifers are generally evergreen is not entirely met, the Japanese larch is present as plantations in the HHNP, being a deciduous conifer. This means that with the proposed method, there is a potential underestimation of conifers, with potential misclassification of the Japanese larch. Thus, it would make sense to include multitemporal information, covering different steps of the phenology of the tree species, including pre-may-sprouting or the peak of the season (Zhu et al., 2008).

With a spatial resolution of 10 m, mixed pixels between deciduous trees and conifers can occur, at the edges between separate pure stands, but more so in mixed forests, which are present in the HHNP. This will mean that a certain amount of misclassification is inevitable. This could be improved by making use of higher resolution auxiliary data, such as high-resolution LiDAR data, or by interpreting sub-pixel fraction abundances (Senf et al., 2020).

6.4. Forest Loss Map

In accordance with the good practices established in Olofsson et al. (2014) the validation and area estimations of a change map, and thus also of a forest loss map, should not be done based on the previously produced forest type maps, but on the forest loss itself, with a separate validation specifically for the forest loss. We have omitted this part due to time constraints, but this results in a very limited interpretability of the forest loss map. An area

adjusted accuracy assessment (Olofsson et al., 2013) and area estimations with margins of error should be done using said validation dataset.

Still, from Fig. 6 and Fig. 7, we can already conclude that the NDVI-based proportional difference seems to be more conservative than the PVI-based one, meaning that more area will be classified as “no change” having occurred. From a visual inspection, some of the conifer areas seem to not be included at all. Some of the area consists of dead standing trees from the previous year (2022), so this would be correctly included in the non-forest area for 2023. But some of the missing conifer areas will certainly point towards inadequacies in the produced non-forest, resulting in a misclassification as deciduous forest. This is supported by the results from the accuracy assessment (Tab. 4), where it is shown that with a PA of $\sim 70 \pm 5 \%$, about $30 \pm 5 \%$ of the coniferous tree area on the ground is misclassified as being deciduous.

The difference in the amount of represented area between the NDVI- and the PVI-based maps is due to the boundary clean, resulting in different amounts of smoothing due to different groups being identified.

6.5. Field Data Collection: Improvements & first results

The **survey design** has not been perfect. Some survey points were not forest at all (e.g. grassland or bogs). Some points were at the edges of the forest stands, between forest and non-forest areas or between different tree species. This leads to time-inefficiency in the survey.

The survey sheet did account for some of these issues, by allowing for a reassignment of the point towards cells in close proximity, in order to guarantee survey points representing pure stands. But the survey design could be improved, by including a removal of cells within known water bodies and a removal of cells within a buffer zone of known streets. From the preselected areas, where the random cells got assigned, a prior step could have been the removal of one row of the outside pixels, to reduce the risk of selecting survey points outside or at the border of the forest stands.

Additionally, multiple maps could have been produced to bring to the field, to give more a priori information to the surveyors, although not possible due to time-constraints. A map including topographic information (e.g. a digital elevation model), would have been useful to assess the steepness and thus difficulty of the terrain. A basic road-network has been included, but this could further encompass known hiking roads. The known water bodies could also be displayed in a separate map.

A lot of survey points were inaccessible, but this seems to be inevitable with the nature of the research question being the spruce vitality and impact of the SBB. In order to optimise the time-management, a prior step in the sampling design could have been to exclude clearly dead tree stands (based on high resolution imagery from 2023), or mark them as such as an additional field in the table.

Including the assignment of inaccessible survey points in the observations, specifying their forest type and a first assessment of their vitality proved useful in the acquisition of a more representative number of observations. This also led to a higher time-efficiency.

In order to improve the survey, the field preparation should include:

A field for the majority tree species should have been included in the survey; without it, the statistical analysis on the tree species is quite cumbersome, as the majority would first need to be estimated from the photos taken during the survey.

Forest cover was assessed visually, with a photo taken from the ground for later verification. However, this method can be subjective and vary depending on the observer. A detailed visual assessment protocol with predefined classes could further reduce this subjectivity. It is also essential to standardise the criteria for diagnosing tree diseases in order to limit subjectivity. In addition, the representativeness and reproducibility of the study could be improved by taking into account the biases associated with the proximity of sampling points to human infrastructures, since these areas are not necessarily representative of the entire forest. However, the proximity of the plots to human infrastructure has been used as a compromise between accessibility, time efficiency and representativeness.

While the **sampling design** was set up in a way to relatively equally sample points for the different natural park zones, this has not entirely been accomplished, with zone 2 being higher represented. In this sense, the survey could have been explicitly directed in a way to heighten the chance for equal samples across the zones, by assigning different groups to different zones specifically. Fig. 9 illustrates how the majority of the conifers are dead standing or in a damaged state. Here, it would be interesting to analyse how many of the healthy conifers are different tree species from the Norway Spruce, but as previously explained, including the majority species in the analysis would be too cumbersome for the context of this preliminary study.

Nonetheless, we can give a first assessment of the impact of the natural zones on the health status of the conifers: Referring to Tab. 5 it can be observed that no p-values for the pairwise two-sided Fisher's exact test are significant, H_0 is retained and we can thus conclude that no impact of the natural park zones on the health status can be observed. However, the reality is not a univariate problem. Multiple factors could potentially influence this outcome, such as the soil conditions and thus water availability or generally stress, which is not covered in our tests. A removal of non-spruce species could also influence the result. Furthermore, although a Fisher's exact test was performed to account for small sample sizes, the sample sizes could still be too small to have a statistical relevance, especially so for the severe damage class with as low as 2 counts for zone 1a. As previously stated, there is some amount of bias in the assignment of different health statuses to the plot, varying in the different groups as well. This will also influence and potentially hide the true distribution of the health statuses for the different zones.

Still, it seems as if the bark beetle outbreak will in long term eliminate most mature spruce in the next few years, leading to a rejuvenation of the forest and potentially to a higher heterogeneity of tree species (Alfaro et al., 2015).

7. Conclusion

This study aimed to assess the impact of the BB infestations on the forest ecosystem in the HHNP, with specific goals being the mapping of forest loss (of conifers and mainly spruce) and the evaluation of the health condition of coniferous tree stands. By making use of a series of Sentinel-2 scenes, forest loss maps were produced based on the NDVI and PVI respectively. Insights into the health status of coniferous trees and its distribution for the different management zones in the national park were gained.

The results show that the bark beetle infestation has contributed to a substantial amount of conifer loss, while a quantification has not been performed due to lack of a validation dataset for the change maps in context of the timeframe of the study. The PVI showed greater sensitivity in detecting changes compared to the NDVI, which suggests that the PVI may be better suited for detecting subtle variations in forest health.

The field data showed that many trees were dead or damaged. No statistically significant relationship was found between tree health and the HHNP management zones. Still a potential influence of different environmental factors, such as water availability or the influence of various stressors, should be kept in mind, which was not covered by this study. This could be a direction for future studies.

Challenges were encountered in the forest type classification and in the field data acquisition. The classification of forest types, especially conifers, was limited by the resolution of the satellite imagery, leading to mixed pixels and in that context also the mixed nature in case of mixed stands. The statistics on the influence of the management zones might not be reliable due to a low sample size for some combinations between health status and management zone (Fig. 9). This can be attributed to logistical issues, by not guaranteeing that all zones are equally covered, although an attempt was made in the sampling design.

The study provides a starting point for further analysis of forest health and on the monitoring of forest loss. Based on the field survey, with some data processing, the study group can have a deeper look into the impact of the BB outbreak on different conifer species, a comparison of field data with satellite data or the influence of different parameters, such as topology, on the health status (during the winter term).

The utility and necessity of repetition of a well-thought-out survey over multiple years is highlighted, especially for the need to validate the forest loss map. But this is of course not feasible in the context of a course.

8. References

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